

4.3 Representing Functions as Power Series

Motivation: Consider the series $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$. We saw earlier that this is a geometric series with interval of convergence: $(-1, 1)$.

Example 1: Find a power series representation for $g(x) = \frac{1}{2+x}$ and state its interval of convergence.

Example 2 (you try): Find a power series representation of the function $f(x) = \frac{x}{2x^2 + 1}$ and state its interval of convergence.

Example 3: Find a power series representation for $g(x) = \frac{1}{(x-1)(x-3)}$. Find its interval of convergence.

Note: Other operations such as multiplication and division can be done with power series, although they are a bit more complicated. We will not discuss these ideas in class, but details of these operations are discussed in the book.

Theorem (Addition & Subtraction of Power Series): Suppose that the power series $\sum_{n=0}^{\infty} c_n (x-x_0)^n$ and $\sum_{n=0}^{\infty} d_n (x-x_0)^n$ converge to the functions f and g , respectively on a common interval I (centered at x_0). Then, the power series $\sum_{n=0}^{\infty} (c_n + d_n) (x-x_0)^n$ converges to the function $f + g$ on I .

Theorem: If the power series $\sum_{n=0}^{\infty} c_n (x - x_0)^n$ has radius of convergence $R > 0$, then the function f defined by $f(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^n$ is differentiable on the interval $(x_0 - R, x_0 + R)$ and

Note: This theorem indicates that we should be able to differentiate a power series infinitely many times!

Example 4: Use $f(x) = \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ on $(-1, 1)$ to express $\frac{1}{(1-x)^2}$ as a power series. State the radius of convergence of the series that you find.

Example 5: Find a power series representation for $g(x) = \ln(1+x)$ and state its radius of convergence.

Example 6 (you try): Find a power series representation for $f(x) = \tan^{-1}(x)$ and state its radius of convergence.